OFFICIAL CRM OPERATING MANUAL • CONFIDENTIAL & PROPRIETARY

MHIRA NATURAL SOAP & COSMETICS

CRM OPERATIONS MANUAL

Standard Operating Procedures across Inventory WIP Batch Conversion, Catalog Merchandising, Customer Storefront Checkout, Logistics Thermal Waybills & Admin Governance

Document Identifier:	`MHIRA-SOP-2026-v1.2`	**Document Version:**	Version 2.0 (Redesigned & Consolidated)
Effective Date:	October 2026	**Security Classification:**	**Confidential & Proprietary**
Target Department Scopes:	Inventory, Marketing, Logistics, Admin	**Primary Custodian:**	Operations & Systems Administration Desk
Technical Architecture:	React 19 SPA • Node / Express • MongoDB	**Regulatory Compliance:**	RA 10173 (Data Privacy Act) • FDA/DTI GMP

OPERATIONAL METRICS & SYSTEM PARAMETERS

- **Closed-Loop Modules:** 5 Operational Phases
- **Governed Procedures:** 15 Standard Procedures
- **Active Staff Accounts:** 4 Department Roles
- **Stock Verification:** Closed-Loop FEFO Tracking
- **Audit Trail Engine:** Immutable Security Logging
- **Disaster Recovery:** Automated Daily Cloud Backups

EXECUTIVE MANDATE & AUDIT ADHERENCE

LEGAL & AUDIT COMPLIANCE NOTICE: This operations manual establishes mandatory operating procedures for the MHIRA Enterprise CRM. All handovers—from inbound raw botanical delivery receiving through automated recipe batch deductions, catalog merchandising, customer checkout, thermal courier waybill printing, and database backups—must adhere to these documented standards. Unauthorized reproduction or dissemination outside authorized MHIRA staff is strictly prohibited under Philippine Republic Act No. 10173.

Document Custodian: Kent Bundalian Operations & Systems Lead	**QA Approval:** Quality Assurance Desk GMP & Formulation Verification	**Executive Sign-off:** Executive Desk MHIRA Natural Soap & Cosmetics

SECTION 0.0 • DOCUMENT CONTROL & SITEMAP

Document Control, Revision History & Table of Contents

This section documents formal document revision history, department approvals, an indexed directory of procedures, and standardized conventions.

0.1 Document Revision & Change Governance

Rev	Effective Date	Author / Custodian	Summary of Operational Changes & Milestone	Status
v1.0	Jan 15, 2026	Systems Team	Initial CRM deployment, basic inventory tracking & supplier directory.	Superseded
v1.1	Jun 20, 2026	Marketing / Ops	Integrated storefront promotions, coupons engine, and loyalty reward tiers.	Superseded
v1.2	Oct 10, 2026	Operations Desk	Implemented automated FEFO expiry monitoring & WIP recipe conversion logic.	Superseded
v2.0	Oct 27, 2026	Kent Bundalian / Ops	Comprehensive document redesign, vector tables, end-to-end SOP checklists, error diagnostics, and administrative governance.	**Official Active**

0.2 Table of Contents & Operations Directory

Section / Module Name	Operational Scope & Key SOP Inclusions	Target Role	Page
Section 1.0 Executive System Overview	Closed-loop operational lifecycle, system architecture & RBAC permission matrix	All Scopes	**Page 3**
Section 2.0 Phase 1: Inventory & Production (Part 1)	Inbound delivery receiving, supplier manifest logging, FEFO lot tracking	Inventory	**Page 4**
Section 2.1 Phase 1: Inventory Management (Part 2)	WIP batch recipe conversion, auto raw material deduction, finished goods handover, expiry audits	Inventory	**Page 5**
Section 3.0 Phase 2: Marketing Merchandising	Products catalog publishing, botanical copy, photo assets, promo vouchers & reviews	Marketing	**Page 6**
Section 4.0 Phase 3: Customer Storefront Experience	Storefront navigation, shopping cart, coupon application, address checkout & loyalty tiers	Customer / Sales	**Page 7**
Section 5.0 Phase 4: Logistics Fulfillment & Dispatch	Order confirmation desk, COD validation, thermal courier waybills, dispatch & returns	Logistics	**Page 8**
Section 6.0 Phase 5: Administration & Governance	Appearance themes, staff account management, security audit logs, maintenance mode & backups	Administrator	**Page 9**
Section 7.0 Troubleshooting & Diagnostics Matrix	SOP error resolution: stock discrepancy, waybill barcode scanning, backup timeouts	All Scopes	**Page 10**
Section 8.0 Operational FAQs & Compliance Sign-off	Operational FAQs, daily verification checklist & formal multi-department sign-off authorization	Operations / QA	**Page 11**

STANDARD DOCUMENT CONVENTIONS & ROLE TAXONOMY

DOCUMENT CONVENTIONS & ROLE TAXONOMY: Bold text indicates mandatory actions or verified UI buttons. Breadcrumbs (e.g. Sidebar > Inventory > Work in Progress) specify exact navigation paths. Department roles are color-coded: **Amber (Inventory)**, **Purple (Marketing)**, **Emerald (Customer)**, **Blue (Logistics)**, and **Rose (Admin Governance)**.

SECTION 1.0 • EXECUTIVE SYSTEM OVERVIEW & LIFECYCLE

System Architecture & The Closed-Loop Operational Lifecycle

The MHIRA Enterprise CRM operates on a sequential, closed-loop operational paradigm. Data recorded in one department automatically populates downstream operational desks, eliminating manual data re-entry, stock discrepancies, and order delays.

1.1 The 5-Phase Operational Lifecycle Flow

Phase 1: Inventory	Phase 2: Marketing	Phase 3: Customer	Phase 4: Logistics	Phase 5: Admin
Production & Conversion Receives raw materials, logs FEFO lots, executes batch recipes with automated raw material deduction into Finished Goods.	**Catalog Merchandising** Assigns retail pricing, uploads studio photos, crafts botanical descriptions, sets promo coupons, and publishes to storefront.	**Storefront Ordering** Customer explores live catalog, reviews benefits, applies promo vouchers, and checks out with validated delivery address.	**Order Confirmation** Logistics validates orders & COD, calculates courier shipping rates, prints thermal waybills, and oversees courier dispatch.	**Governance & Audit** Manages staff credentials, oversees RBAC access scopes, monitors immutable audit logs, and runs cloud disaster backups.
Output: Stocked Finished Goods	**Output: Live Storefront Catalog**	**Output: Confirmed Placed Order**	**Output: Delivered Parcel & Proof**	**Output: Secure Audit & Backups**

1.2 Role-Based Access Control (RBAC) Permissions Matrix

Functional Module	Admin Role	Inventory Role	Marketing Role	Logistics Role
Raw Materials & Delivery Receiving	View Only	**Full Manage**	No Access	No Access
Work-in-Progress (WIP) Conversion	Audit Only	**Full Manage**	No Access	No Access
Finished Goods Stock Records	View Only	**Full Manage**	View Only	View Only
Products Catalog & Store Merchandising	Audit Only	No Access	**Full Manage**	No Access
Promo Vouchers, Content & Reviews	Audit Only	No Access	**Full Manage**	No Access
Order Confirmation Desk & Dispatch	Audit Only	No Access	No Access	**Full Manage**
Customer Support & Returns Management	Audit Only	No Access	No Access	**Full Manage**
User Governance, System Maintenance & Backups	**Full Control**	No Access	No Access	No Access

1.3 Navigation Hierarchy & Authentication Security

PLATFORM SPECIFICATIONS & RA 10173 COMPLIANCE

SIDEBAR NAVIGATION STRUCTURE: Menu options are strictly role-filtered into four tiers: **Dashboard** (KPI operational snapshot), **Records** (primary data directories), **Process** (active execution workflows), and **Settings** (department reports and configuration). Staff only see authorized modules for their role.

AUTHENTICATION & RA 10173 COMPLIANCE: User logins issue cryptographically signed JSON Web Tokens (JWT) stored in secure cookies. Sessions auto-expire after 8 hours of inactivity. Customer personal identifying data (PII) is masked in logs and restricted strictly to logistics.

PHASE 01 • INVENTORY & PRODUCTION MANAGEMENT (PART 1)

Raw Materials Inbound Receiving & FEFO Lot Management

The Inventory module governs the raw material lifecycle—from physical supplier delivery receiving through First-Expired, First-Out (FEFO) batch logging and stock acceptance.

1.1 Inbound Raw Materials Stock-In & Supplier Receiving

Navigation: Sidebar > Inventory > Delivery Receiving

ROLE: INVENTORY STAFF

Step-by-Step SOP Protocol:

1. **Physical Receiving & Inspection:** Inspect incoming delivery shipments against the supplier Purchase Order (PO). Verify physical seals, package integrity, and attached Certificates of Analysis (CoA).
2. **Manifest Logging:** Open Delivery Receiving and log the supplier name, PO number, internal item code, and carrier delivery receipt (DR).
3. **FEFO Lot & Expiry Registration:** Record the manufacturer batch lot code and strict expiration date. The CRM automatically prioritizes earlier-expiring lots during production conversions.
4. **Stock Acceptance:** Verify received weight (kg) against invoice. Click **Confirm Stock-In** to automatically increment warehouse inventory balances and generate receiving manifests.

Item Code	Raw Material Description & Botanical Grade	Batch Lot No.	Inbound Qty	FEFO Expiry Date	Status
RM0027	**Pure Virgin Coconut Oil** Cold-Pressed Botanical Grade (100% Organic)	LOT-CO-2026-04	**+150.00 kg**	October 15, 2028	**Received**
RM0034	**Sodium Hydroxide (Lye)** Technical Grade 99% Saponifier Pellets	LOT-LYE-2026-02	**+50.00 kg**	May 20, 2029	**Received**
RM0036	**Pure Kojic Dipalmitate** Skin Brightening Active Powder (Pharmaceutical)	LOT-KOJIC-2026-01	**+25.00 kg**	December 10, 2027	**Received**
RM0024	**Papaya Fruit Enzyme Extract** Cold-Extracted Bio-Active Exfoliant Liquid	LOT-PAP-2026-03	**+30.00 kg**	September 04, 2027	**Received**

1.2 Work-in-Progress (WIP) Batch Conversion & Recipe Auto-Deduction

Navigation: Sidebar > Inventory > Work in Progress (WIP)

ROLE: INVENTORY STAFF

Technical Conversion Protocol:

1. **Formulation Selection:** In the WIP dashboard, select the target product recipe (e.g. *Gluta Kojic Soap Formula*). The system automatically retrieves verified master ingredient ratios.
2. **Batch Sizing:** Input the target output quantity (e.g. 100 Finished Bars @ 135g). The engine computes exact raw material consumption weights.
3. **Automated Raw Material Stock Deduction:** Upon clicking **Convert >> Finished Goods**, the system executes atomic database transactions: raw material inventories are decremented, and finished goods are incremented simultaneously with complete audit trails.

CRITICAL QUALITY ASSURANCE & FEFO ALLOCATION MANDATE

FEFO ALLOCATION ALGORITHM: The CRM strictly enforces First-Expired, First-Out picking logic. If an inventory officer attempts to convert a newer lot while an older unexpired lot exists in warehouse stock, the system generates a priority prompt. Raw materials expiring within 30 days are automatically placed in Quarantine and blocked from production batches.

PHASE 01 • INVENTORY & PRODUCTION MANAGEMENT (PART 2)

WIP Conversion Engine & Finished Goods Handover

This section illustrates the actual Batch Conversion Modal UI, automated raw material deduction preview, finished goods warehouse handover, and auxiliary inventory protocols.

FRONTEND VIEW: BATCH PRODUCTION & CONVERSION ENGINE	AUTO-DEDUCTION ENGINE
Selected Formulation: FACE SERUM COLLAGEN Formula (Standard Batch: 5.00 kg) **Target Output:** **166 Finished Bottles (30 ML net wt)**	**Lot Number:** LOT-FG-2026-10-04 **Production Line:** Cleanroom Cosmetic Lab 02
AUTOMATED RAW MATERIAL STOCK DEDUCTION BREAKDOWN (ATOMIC DEDUCTION PREVIEW):	
• Deionized Water (RM0001): **-4,540 g (-4.54 kg)** • Sodium Hyaluronate C (RM0002): **-100 g (-0.10 kg)**	• Botanical Extracts (Tomato, Witch Hazel, Cucumber): **-150 g (-0.15 kg)** • Actives (Bust-A, Azelaic Optima, Preservatives): **-210 g (-0.21 kg)**
✓ All 12 raw material balances verified. Stocks sufficient for 5.00 kg batch execution.	**[ACTION: Convert >> Finished Goods]**

1.3 Finished Goods Stock Handover & Warehouse Masterlist

Navigation: Sidebar > Inventory > Finished Goods

ROLE: INVENTORY STAFF

Finished Goods Handover Protocol: Once batch conversion completes, units are instantly posted into Finished Goods with unique finished SKU, lot number, and FEFO expiry tracking. Quality inspection verifies fill volume and packaging seal before releasing stock to Marketing.

CODE	IMAGE	CATEGORY	ITEM NAME	QUANTITY & CONDITION	SPECIFICATION & UOM	REQUIRED INGREDIENTS (5KG STANDARD BATCH)	ACTION
FG0004		Skin Care	**SERUM** FACE SERUM COLLAGEN	**18 Units** ● Good Condition	**30 ML** 5 kg Batch	**12 Actives (5,000g Batch):** Water 4540g • Hyaluronate C 100g • Tomato 50g • Witch Hazel 50g • Cucumber 50g • Bust-A 40g • Azelaic 40g • Phenoxy 30g • Benzoate 25g • Sorbate 25g • Pore-MinEx 25g • Scent 25g	[View BOM] [WIP Log]

1.4 Auxiliary Inventory SOPs: Expiry Audits & Purchase Requests

SOP Procedure	System Navigation Path	Trigger & Operational Protocol
SOP 1.4: FEFO Expiry Audits	`Sidebar > Inventory > Expiry Monitoring`	Routine automated scan for ingredients reaching 90-day (Warning) and 30-day (Quarantine) thresholds. Quarantined stock requires physical segregation and QA sign-off.
SOP 1.5: Purchase Request	`Sidebar > Inventory > Purchase Request`	Triggered automatically when raw material balances fall below Safety Stock thresholds. Generates requisition tickets for management procurement approval.
SOP 1.6: Supplier Evaluation	`Sidebar > Inventory > Suppliers`	Maintains audited supplier registry, contact personnel, payment terms, and delivery lead times to ensure zero supply-chain downtime.

INTER-DEPARTMENTAL HANDOVER GATEWAY (INVENTORY -> MARKETING)

CLOSED-LOOP HANDOVER VERIFICATION: Finished goods stock levels strictly govern storefront purchasing limits. Marketing cannot publish storefront stock beyond physical Finished Goods counts, guaranteeing zero overselling.

PHASE 02 • MARKETING MANAGEMENT & MERCHANDISING

Product Merchandising, Storefront Publishing & Campaigns

The Marketing department bridges production with customer demand. Merchandisers configure retail prices, upload professional photo assets, craft botanical benefit descriptions, and configure promo campaigns.

2.1 Storefront Product Merchandising: Pricing, Imagery & Copy

Navigation: Sidebar > Marketing > Products Catalog

ROLE: MARKETING STAFF

Merchandising Protocol:

1. **Product Selection:** Select stocked Finished Goods from the catalog table.
2. **Retail Pricing:** Input unit selling price in Philippine Pesos (PHP). Base price must comply with executive margin guidelines.
3. **Botanical Descriptions:** Write FDA-compliant botanical ingredient copy highlighting organic cold-process benefits.
4. **Studio Photography:** Upload high-resolution studio photo assets (aspect ratio 1:1, max 2MB, WebP/PNG).
5. **Publishing Toggle:** Switch status to **Published** to push the item to the live storefront collection.



FACE SERUM COLLAGEN
✓ Live Studio Photo Asset

FRONTEND VIEW: MARKETING PRODUCT MERCHANDISING MODAL

Product Name:	**FACE SERUM COLLAGEN (30 ML) [SERUM]**
Selling Price:	**PHP 300.00** (SKU: FG0004 • Skin Care)
Botanical Benefits:	Artisanal collagen serum formulated with active Sodium Hyaluronate, organic Tomato & Cucumber extracts, Witch Hazel, and Azelaic Optima for skin elasticity and luminous glow.
Storefront Status:	● **Published & Visible to Customers**

2.2 Marketing Campaign Suite: Coupons, Feedback, Content & Events

Navigation: Sidebar > Marketing > Campaign Suite

ROLE: MARKETING STAFF

Campaign Module	System Navigation Path	Marketing Procedure & Governance Rules
Products Vouchers	`Sidebar > Marketing > Products Vouchers`	Generate promo codes (e.g. NATURAL20 for 20% off). Configure minimum spend criteria (PHP 300), total redemption limits, and active date windows.
Customer Feedback	`Sidebar > Marketing > Customer Feedback`	Audit verified purchaser reviews and star ratings. Approve legitimate testimonials for storefront publication; flag suspicious reviews for moderation.
Marketing Content	`Sidebar > Marketing > Marketing Content`	Maintain homepage hero carousels, seasonal promotional banners, and educational botanical skincare guidance articles.
Events Management	`Sidebar > Marketing > Events Management`	Publish event schedules for organic trade exhibitions, wellness pop-up bazaars, and seasonal holiday promotions.
Our Story & Heritage	`Sidebar > Marketing > Our Story Management`	Maintain brand heritage biography, DTI / FDA manufacturing certifications, and corporate values for public storefront display.
Products Promotions	`Sidebar > Marketing > Products Promotions`	Configure bundled promotional packages (e.g. soap & herbal tea gift boxes) with specialized bundle pricing rules.

PHASE 03 • CUSTOMER STOREFRONT EXPERIENCE

Browsing, Checkout Drawer & Customer Loyalty Tiers

The customer storefront is the public e-commerce gateway. Customers discover handcrafted products, validate promotional coupons, select verified shipping addresses, and checkout using diverse payment channels.

3.1 - 3.2 Storefront Exploration, Cart Checkout & Voucher Discount

Navigation: Public URL /collection and /checkout

ROLE: CUSTOMER / SHOPPER

Customer Checkout Flow:

1. **Catalog Exploration:** Shoppers browse /collection, filter by botanical category, and view ingredient transparency.
2. **Cart Drawer:** Items are added to the slide-over cart drawer with real-time stock availability verification.
3. **Voucher Code Application:** Customers apply promo codes (e.g. NATURAL20). The cart engine immediately validates minimum spend rules and subtracts the exact discount.
4. **Philippine Address Validation & Shipping:** Customer selects Region, Province, City/Municipality, and Barangay. Shipping rates compute automatically based on parcel weight and delivery destination.
5. **Payment Options:** Shopper chooses **Cash on Delivery (COD)**, **GCash**, **Maya**, or Credit/Debit Card. Clicking **Place Order** generates order record in status Order Placed.

FRONTEND VIEW: CUSTOMER CHECKOUT DRAWER & ORDER SUMMARY	STATUS: ORDER PLACED
Purchased Items: 1x FACE SERUM COLLAGEN (30 ML) [SERUM] SKU: FG0004 • Unit Price: PHP 300.00	**Item Subtotal:** **PHP 300.00**
Promotional Voucher Applied: NATURAL20 (20% Off)	**-PHP 60.00**
Standard Courier Shipping: Cabuyao, Laguna (CALABARZON)	**PHP 65.00**
TOTAL PAYABLE AMOUNT:	**PHP 305.00**
Selected Payment Method: Cash on Delivery (COD Collect)	**✓ Awaiting Logistics Confirmation**

3.3 Order Tracking & Customer Loyalty Rewards Program

Navigation: Storefront /track-order & Profile

ROLE: CUSTOMER / SHOPPER

Customer Loyalty Rewards Engine:

Customers earn reward points automatically upon order delivery: **1 Reward Point per PHP 50.00 spent** on confirmed delivered orders. Accumulated points unlock progressive VIP tiers:

Loyalty Tier	Spending Threshold	Point Multiplier	Exclusive Customer Benefits & Storefront Privileges
Bronze	Under PHP 1,000	1.0x (Standard)	Standard loyalty points accumulation, order status tracking, birthday discount code.
Silver	PHP 1,000 – PHP 4,999	1.2x Multiplier	Free shipping on orders above PHP 500, priority customer support ticket resolution.
Gold	PHP 5,000 – PHP 9,999	1.5x Multiplier	5% permanent account discount, early access to limited seasonal botanical releases.
Platinum VIP	PHP 10,000 and above	2.0x Multiplier	10% permanent account discount, complimentary gift packaging, direct VIP hotline.

PHASE 04 • LOGISTICS FULFILLMENT & DISPATCH

Order Confirmation Desk, Thermal Waybills & Courier Dispatch

Logistics serves as the official Order Confirmation authority. The desk verifies customer addresses, validates COD phone numbers, generates thermal waybill labels, and coordinates courier handover.

4.1 Customer Order Confirmation Desk & Address Verification

Navigation: Sidebar > Logistics > Order Management

ROLE: LOGISTICS STAFF

Order Confirmation Protocol:

1. **Order Triage:** Review incoming orders in status Order Placed.
2. **Address & Contact Verification:** Confirm Philippine barangay, municipality, and 11-digit mobile phone. For COD orders above PHP 1,000, call or SMS recipient before confirmation.
3. **Status Confirmation:** Click **Confirm Order** to transition record from Order Placed into Processing. Finished goods are immediately reserved for parcel packing.

4.2 Thermal Waybill Printing & Courier Dispatch

Navigation: Sidebar > Logistics > Order Management

ROLE: LOGISTICS STAFF

MHIRA LOGISTICS EXPRESS	TRACKING: #JNT-PH-00293182
COURIER: J&T; EXPRESS PHILIPPINES (STANDARD AIRWAY BILL)	DATE: 2026-10-27
JNT-PH-00293182	
SENDER: Mhira Natural Soap & Cosmetics Brgy. Sala, Cabuyao, Laguna 4025 Contact: 0917-123-4567	**RECIPIENT:** Kent Bundalian Brgy. Sala, Cabuyao, Laguna 4025 Contact: 0906-525-8847
PARCEL CONTENTS: 1x FACE SERUM COLLAGEN (30 ML) [FG0004] **TOTAL WEIGHT:** 0.15 kg • **DIMENSIONS:** 12cm x 6cm x 6cm	**PAYMENT METHOD:** COD COLLECT: PHP 305.00
✓ Order Confirmed by Logistics Desk	[Print Waybill] [Dispatch (In Transit)]

4.3 - 4.4 Courier Handover, In-Transit Tracking & Returns Management

Navigation: Sidebar > Logistics > Return Management

ROLE: LOGISTICS STAFF

Delivery Lifecycle & Returns Governance:

1. **Courier Handover:** When courier rider receives parcels, update status to In Transit.
2. **Delivery Handover:** Upon electronic proof-of-delivery (e-POD), mark order Delivered. This automatically updates customer loyalty points.
3. **Returns & Exchange SOP:** In Return Management, verify return requests within 7 days of delivery. For damaged in-transit goods, file courier claims. Returned soaps cannot be restocked for hygiene safety; write off to Scrap.

PHASE 05 • ADMIN GOVERNANCE, SECURITY & DISASTER RECOVERY

Appearance, Staff Accounts, Audit Trails & Cloud Backups

The Administration role governs system integrity, security policies, role permissions, emergency maintenance mode, and database backup snapshots.

5.1 - 5.2 Appearance Customization & Staff User Access Control

Navigation: Sidebar > Admin > User Management

ROLE: SYSTEM ADMIN

User Administration Protocol:

1. **Account Provisioning:** Create staff accounts, specify email, username, and assigned department role.
2. **Credential Policy:** Passwords require minimum 8 characters with alphanumeric and special characters. Enforce credential reset upon first login.
3. **Deactivation Protocol:** Former employee accounts must be deactivated immediately to revoke token access.

Frontend View: Admin > User Management Directory

Staff Member & Handle	Email Address	Assigned Role	Status	System Access Scope & Department Responsibilities
Administrator @admin	admin@mhiracosmetics.ph	Admin	Active	Full governance, user administration, maintenance mode & backups.
Kent Bundalian @logistics_kent	logistics@mhiracosmetics.ph	Logistics	Active	Order confirmation desk, thermal waybill printing, courier dispatch.
Inventory Officer @inv_officer	inventory@mhiracosmetics.ph	Inventory	Active	Raw material receiving, FEFO expiry tracking, WIP recipe conversion.
Marketing Lead @marketing_team	marketing@mhiracosmetics.ph	Marketing	Active	Products catalog merchandising, promotional vouchers, customer feedback.

5.3 - 5.5 Security Audit Trails, Emergency Maintenance & Cloud Backups

Navigation: Sidebar > Admin > Maintenance & Backup

ROLE: SYSTEM ADMIN

Governance Pillar	System Navigation Path	Technical Protocol & Disaster Recovery Rules
SOP 5.3: Security Audit	`Sidebar > Admin > User Activity`	Monitors immutable security logs: login timestamps, IP addresses, batch production executions, and status modifications. Records cannot be purged.
SOP 5.4: Maintenance Mode	`Sidebar > Admin > System Maintenance`	Locks customer storefront and displays polite maintenance notice during database migrations or updates. Admin portal remains accessible.
SOP 5.5: Database Backup	`Sidebar > Admin > Data Backup`	Generates automated daily JSON / MongoDB snapshots. Administrator can trigger on-demand backups before production recipe or catalog imports.

SECTION 7.0 • SYSTEM DIAGNOSTICS & TROUBLESHOOTING

Incident Diagnostics & Operational Troubleshooting Matrix

This section outlines standard incident triage protocols, root-cause analyses, and step-by-step resolution procedures for common operational errors across inventory, marketing, checkout, logistics, and database administration.

Operational Symptom / Error Message	Root Cause Analysis	Role Scope	Step-by-Step Resolution Procedure
WIP Conversion Error: ***'Insufficient raw material balance for formulation'***	Warehouse raw material stock is lower than required recipe quantity, or available lots are in Quarantine.	Inventory	1. Check Delivery Receiving for pending shipments. 2. Verify if lots are locked in Expiry Monitoring quarantine. 3. Stock-in fresh materials before re-attempting conversion.
Storefront Discrepancy: ***Product displays 'Out of Stock' despite active listing***	Finished goods physical stock count reached 0. Storefront availability strictly mirrors inventory balances.	Marketing / Inventory	1. Contact Inventory Officer to execute a fresh WIP batch. 2. Once batch converts to Finished Goods, storefront status restores automatically without manual re-publishing.
Order Confirmation Delay: ***Logistics unable to transition order from 'Order Placed'***	Recipient mobile number is invalid or delivery address lacks required Philippine barangay data.	Logistics	1. Open Order Details modal in Order Management. 2. Contact customer via email/phone to verify complete address. 3. Edit address fields, click **Save**, and confirm order.
Thermal Waybill Barcode Error: ***Courier rider unable to scan printed barcode***	Thermal printer density too low, thermal paper creased, or browser print scale distorted.	Logistics	1. Clean thermal printer printhead with isopropyl alcohol. 2. In print dialog, ensure scale is 100% (do not fit to printable area). 3. Reprint official waybill label from Order Management.
Access Denied / 403 Forbidden: ***Staff member cannot access specific sidebar module***	Staff account does not possess the required RBAC role permissions, or session token expired.	Admin / Staff	1. Log out and log back in to refresh JWT session. 2. If still restricted, Admin must check User Management to confirm assigned role scope matches job description.
Database Backup Warning: ***'Snapshot generation timeout or disk write limit'***	Server storage quota reached, or heavy read/write concurrency during peak hours.	Admin	1. Archive older backup snapshots to offsite cloud storage. 2. Trigger on-demand backup during non-peak operational windows (after 10:00 PM). 3. Verify MongoDB disk health in server cPanel.

CRITICAL INCIDENT ESCALATION & AUDIT SAFEGUARD

INCIDENT ESCALATION PROTOCOL: If an operational error cannot be resolved through standard diagnostics within 15 minutes, escalate immediately to the Systems Administrator via admin@mhiracosmetics.ph or the emergency support channel. Do not attempt direct database record alterations without written authorization from the Operations Lead.

SECTION 8.0 • OPERATIONAL FAQS & DOCUMENT AUTHORIZATION

Operational FAQs, Audit Checklist & Sign-Off Authorization

This concluding section addresses core operational inquiries, establishes a daily/weekly readiness checklist, and formalizes departmental compliance through official executive signatures.

8.1 Frequently Asked Questions (FAQs)

Q: How does raw material auto-deduction prevent inventory discrepancies? A: When an Inventory Officer converts a batch in WIP, the formula engine mathematically computes exact ingredient weights and executes atomic database deductions. This guarantees zero unrecorded raw material consumption.	**Q: Can Marketing publish storefront products without physical finished goods?** A: Marketing can draft descriptions and upload imagery, but storefront availability strictly mirrors verified Finished Goods inventory counts to prevent overselling.
Q: Why does Logistics act as the sole Order Confirmation authority? A: Logistics verifies Philippine regional addresses, checks parcel weight against courier shipping rate tiers, validates COD contact legitimacy, and prints official thermal waybills before courier handover.	**Q: How are customer loyalty tier reward points calculated?** A: Customers automatically earn 1 Reward Point per PHP 50.00 spent on delivered orders. Spending thresholds: Bronze (<PHP 1,000), Silver (PHP 1,000+), Gold (PHP 5,000+), and Platinum VIP (PHP 10,000+).

8.2 Daily & Weekly Operational Readiness Checklist

Frequency	Operational Inspection Item	Responsible Role	Verification Standard & Criteria
Daily (AM)	**Inbound Manifest & FEFO Audit**	Inventory	Inspect physical stock-in, verify CoA, check 90-day expiry monitoring alerts.
Daily (AM/PM)	**Order Confirmation & Dispatch Triage**	Logistics	Confirm all Order Placed records, print thermal waybills, dispatch riders.
Daily (PM)	**Storefront Stock & Reviews Audit**	Marketing	Review out-of-stock items, approve pending verified customer product reviews.
Daily (Midnight)	**Automated Database Cloud Backup**	Admin	Verify automated daily snapshot creation and log zero database sync errors.
Weekly	**Physical Stock vs System Variance Audit**	Inventory / Ops	Conduct physical cycle count on raw botanical oils and finished soaps; zero tolerance for unrecorded variance.

8.3 Document Authorization & Compliance Sign-Off

PREPARED BY:	REVIEWED BY:	APPROVED BY:
Kent Bundalian Systems Architect & Documentation Specialist Date: October 27, 2026	Head of Formulations & QA Quality Assurance & GMP Compliance Date: October 27, 2026	Director of Operations MHIRA Natural Soap & Cosmetics Date: October 27, 2026